ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.35	**Gender**	Male
Height (cm)	175.0	**Weight (kg)**	70.0
BMI	22.9	**Product Info 2**	A1
Occupation	Software Developer	**Annual Income**	Rs. 120,000.00
Insurance Type	Life	**Coverage Amount**	Rs. 500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	**Low Risk**
Recommended Annual Premium	**Rs. 10,000.00**

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Applicant: customer1
Insurance Type: Life
Coverage Amount: Rs. 500,000.00

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1. Risk Assessment

Based on the provided applicant details, a preliminary risk assessment reveals several factors:

* **Age:** 0.35 years (approximately 4 months). This is an extremely young age.
* **Health & Lifestyle:**
* **BMI:** 22.9, which falls within a healthy range.
* **Smoker:** Non-smoker (0).
* **Alcohol:** Abstinent (0).
* **Exercise:** Active (1).
* **Financial & Occupational:**
* **Occupation:** Software Developer. This is generally considered a low-risk occupation in terms of physical hazard.
* **Income:** Rs. 120,000.00 per annum.
* **Insurance History:**
* **Previous Claims:** 0 (no previous claims).
* **Family History:** 0 (no adverse family history indicated).
* **Product Information:** 'A1' (specific product details not provided but assumed to be relevant to the ML model).
* **Gender:** Male.

Initial Assessment: Excluding the age, the applicant's profile (BMI, non-smoker, abstinent, active lifestyle, low-risk occupation, no adverse history) suggests a very low-risk individual. However, the reported age of 0.35 years presents a critical inconsistency when combined with other attributes (see Anomaly Detection).

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2. Comparative Analysis

The current application is compared against the top three similar historical cases:

1. **Application APP-3D4F9D (Similarity: 97.1%):**
* **Profile:** Risk Class 1, Premium Rs. 10,000.00, Status: **rejected**.
* **Comparison:** This case is highly similar to the current application, matching the predicted Risk Class (1) and Premium (Rs. 10,000.00). **Crucially, despite the high similarity and low-risk profile, this application was rejected.** This suggests there might be an underlying underwriting rule or policy exclusion (e.g., minimum age, specific product eligibility, or other non-quantifiable factors) that the ML model does not explicitly capture, which led to rejection. The client email "officer@shield.com" might indicate a corporate or specific group context,

potentially with unique underwriting considerations.

2. **Application POL-4519 (Client: customer1, Similarity: 94.5%):**
* **Profile:** Risk Class 4, Premium Rs. 17,000.00, Status: **approved**.
* **Comparison:** This is an approved application for the same client ('customer1'). However, it had a significantly higher Risk Class (4) and a higher premium (Rs. 17,000.00) compared to the current application. This indicates that 'customer1' has previously been approved for insurance, but likely under different circumstances (e.g., at a different age, for a different product, or with a different risk profile at that time).

3. **Application POL-8902 (Client: customer1, Similarity: 94.0%):**
* **Profile:** Risk Class 2, Premium Rs. 12,000.00, Status: **approved**.
* **Comparison:** Another approved application for 'customer1', with a higher Risk Class (2) and premium (Rs. 12,000.00) than the current application. Similar to POL-4519, this suggests 'customer1' is an existing customer but the current application's profile (as assessed by the ML model) is much more favorable.

Summary: The current application presents as a very low-risk case (Risk Class 1). While the 'customer1' client has approved policies with higher risk classes, the rejection of a nearly identical historical case (APP-3D4F9D) is a significant red flag that warrants further investigation, especially given the identified anomaly.

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3. Anomaly & Fraud Detection

Critical Anomaly Detected:
There is a severe data inconsistency in the applicant's profile:
* **Age:** 0.35 years (approximately 4 months old).
* **Physical Attributes:** Height 175.0 cm, Weight 70.0 kg, BMI 22.9. These are typical adult measurements. A 4-month-old infant would be significantly smaller (e.g., height typically 60-70cm, weight 6-9kg).
* **Occupation & Income:** "Software Developer" with an income of Rs. 120,000.00. An infant cannot hold an occupation or earn income.

Conclusion: The provided data is fundamentally contradictory. It is highly probable that either:
1. The `age` field refers to the insured child, while `height`, `weight`, `occupation`, and `income` refer to the policyholder/parent.
2. The `age` field is incorrect, and the applicant is an adult whose physical/occupational details are correctly stated.
3. There is a significant data entry error or data corruption.

This anomaly severely impacts the reliability of the ML classifier's output and any subsequent underwriting decision without clarification.

Other potential anomalies:
* **Coverage vs. Income:** If the stated income (Rs. 120,000.00) is truly that of a 0.35-year-old (which is impossible), then a coverage amount of Rs. 500,000.00 (approximately 4.17 times income) would be highly unusual and potentially indicative of over-insurance. However, if the income belongs to a parent and the policy is for a child, this ratio would need to be assessed against the parent's income and the purpose of the child's policy.

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4. Premium Recommendation

The Machine Learning Classifier has calculated an Annual Premium of **Rs. 10,000.00** for a Predicted Risk Class of 1, with a confidence of 73.2%.

* **Justification (based on ML output, assuming data validity):** The premium is justified by the extremely low predicted risk class (1/8), indicating a highly favorable risk profile based on the provided attributes (healthy BMI, non-smoker, active, no claims, no family history, low-risk occupation).
* **Comparison with RAG:** The premium of Rs. 10,000.00 aligns with the highly similar (97.1%) historical case APP-3D4F9D, which also had a Risk Class of 1 and the same premium.
* **Caveat:** The validity of this premium recommendation is entirely dependent on resolving the critical data anomaly. If the applicant is indeed an infant, the current ML output (based on adult physical and occupational attributes) is irrelevant and the premium would need to be re-evaluated under appropriate child policy terms. If the applicant is an adult, the age field is incorrect, and the premium might be appropriate given the other favorable attributes.

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5. Underwriting Decision Recommendation

Recommendation: Refer for Manual Review

Due to the **critical and irreconcilable data inconsistency** between the applicant's age (0.35 years) and their reported height, weight, occupation, and income, a direct automated decision to approve or reject is not possible and would be irresponsible.

Actions Required during Manual Review:

1. **Clarify Applicant Identity and Age:** Immediately investigate and confirm the actual age and identity of the applicant.
* Is the applicant truly 0.35 years old? If so, the height, weight, occupation, and income details are erroneous and must be corrected/removed.
* If the applicant is an adult, the `age` field is incorrect and needs to be updated.
* Clarify if this is a policy for a child (the 0.35-year-old) where the stated occupation and income belong to the parent/policyholder.

2. **Re-evaluate Risk Profile:** Once the data is clarified and corrected, the applicant's risk profile must be re-evaluated. If the applicant is an infant, specific child-focused underwriting guidelines and product eligibility will apply, and the current ML prediction based on adult attributes will be invalid.

3. **Investigate Similar Rejection (APP-3D4F9D):** During manual review, investigate the reasons for the rejection of APP-3D4F9D. Given its high similarity to the current application (in terms of ML output), understanding its rejection criteria is vital to ensure no hidden underwriting rules or policy exclusions are being triggered.

4. **Premium Reassessment:** Based on the verified applicant data and re-evaluated risk profile, a new premium calculation may be necessary.

This application cannot proceed without manual intervention to resolve the fundamental data discrepancies.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **Rs. 10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.